

Decomposing Allen–Dynes error into learnable and irreducible components

Abhinav Sharma*

Independent researcher, Mumbai, India

Abstract

The Allen–Dynes formula predicts the superconducting transition temperature from three moments of the electron–phonon spectral function $\alpha^2F(\omega)$, and underpins essentially all large-scale computational searches for new superconductors. We ask what that compression costs, separating error a better function of the same three moments could absorb from error no such function can, because the information was destroyed when α^2F was reduced to three numbers.

Solving the linearised isotropic Migdal–Eliashberg equations for 806 DFPT spectral functions, we find both components governed by the same spectral shape parameter $r = \omega_2/\omega_{\log}$, and distinguishable by construction: the learnable component is variation predictable from r , an input the formula already receives, while the irreducible component is the T_c spread among spectra sharing all three moments, which no such function can recover. The irreducible floor rises steeply with r ($\rho = +0.92$ over nine moment-matched ensembles); measured at fixed r it is non-monotone in the coupling constant, with a minimum near $\lambda = 0.9$ and a maximum near 5.

Relating the two requires declaring the recalibration class, the population and the μ^* convention. Under our declarations Allen–Dynes sits a factor 11.0 above the floor, and between 3.3 and 20.6 across every alternative we tabulate; correcting for the coupling coverage of the moment-matched cells brings it to 5–11, the width of that interval being set by the estimand rather than by the measurement. The order of magnitude is robust, the leading digit is not. Matched in shape space, headroom falls from 26.7 at $r = 1.06$ to 2.9 at $r = 1.56$ ($\rho = -0.833$), because the floor rises with r far faster than the error does. Where the spectrum is near-Einstein a better closed form has an order of magnitude to give; where it is broad, three moments are already close to exhausted.

Throughout, the reference T_c is the Migdal–Eliashberg solution on DFT input, not the experimental value: this work compares a closed-form approximation with the theory it approximates, on identical input.

1. Introduction

Predicting the superconducting transition temperature of a candidate material from first principles requires solving the Migdal–Eliashberg equations, whose input is not a number but a function: the electron–phonon spectral function $\alpha^2F(\omega)$. Because that solution is expensive, essentially all large-scale computational searches instead evaluate a closed form — the Allen–Dynes modification of McMillan’s equation [McMillan 1968; Allen & Dynes 1975] — which takes three moments of α^2F and returns T_c directly. Machine-learned T_c predictors are built on the same three descriptors.

That substitution has a cost, and the cost has two parts that are usually not separated:

- a **learnable** part, arising because the closed form is an interpolation fit and handles its own three arguments imperfectly. A better function of the same arguments — analytic or learned — can absorb it.
- an **irreducible** part, arising because infinitely many distinct α^2F share the same three moments and have different true T_c . That information is destroyed at the moment of compression. No predictor taking only those three inputs can recover it, however much data it is trained on.

*Correspondence: abhinavsharma1359@gmail.com

The distinction matters for how effort is allocated. If the irreducible part is small, the closed form is near-optimal on its own inputs and improvement should be sought in better descriptors or better DFT. If it is large, the field is pursuing a precision that does not exist in the representation, and a fourth descriptor is required before any model can improve.

Relation to existing work

Two components of this question are already addressed in the literature and we claim neither.

The **cutoff convention for the Coulomb pseudopotential** — that μ^* has no meaning without the energy scale at which it is defined, and that comparisons between Eliashberg and Allen–Dynes results are invalid unless it is matched — is derived and applied by Kogler *et al.* [IsoME, 2025], who conclude that “the previously held assumption that Eliashberg theory overestimates T_c is no longer valid when μ^* is appropriately adjusted.” We reach the same convention independently and use the agreement as a check on our solver (§3.3), not as a contribution.

Improving the closed form by regression on its own moments is the subject of Xie *et al.* [2022], who applied symbolic regression and SISO to $(\lambda, \mu^*, \omega_{\log}, \bar{\omega}_2)$ against numerically solved Eliashberg T_c . On their non-hydride set the Allen–Dynes formula gives a 6.5% RMSE against their fit’s 7.3%; on hydrides the figures are 18.7% and 5.8%, and on artificial Gaussian spectra 27.3% and 9.2%. That improvement is the learnable component of the decomposition above, measured on a different population from ours: their artificial spectra reach $\lambda \approx 4$ and their materials are weighted toward hydrides, so the percentages are not comparable with ours term-by-term.

The **moment-matched set itself** is not new either, and has a long history in its extremal form. Bergmann and Rainer [1973] computed the functional derivative $\delta T_c / \delta \alpha^2 F$ and found every frequency region to contribute positively, most strongly just above $2\pi T_c$; and the optimal spectrum under a fixed coupling constant is a narrow peak at the highest available frequency. Gao *et al.* [2025] make that last statement quantitative in the course of bounding the maximum T_c of ambient-pressure conventional superconductors, and observe that the optimum is physically unattainable because λ and $\omega_{\log}$ are derived from the same spectral function and trade against each other. All of these are statements about the extreme point of the set we study.

Shape dependence has been measured before, and non-monotonically. Akis and Carbotte [1989] varied model spectral shapes at fixed strong-coupling index $T_c / \omega_{\ln}$ and found that “the degree of shape dependence of each property is sensitive to the value of $T_c / \omega_{\ln}$ ”, and — the sentence that matters here — that “the amount of shape dependence does not increase steadily with $T_c / \omega_{\ln}$ and there appears to be regions of maximum shape sensitivity.” That is the qualitative structure of §4.9, found in 1989 by a different method. Converting our rungs to their index, our measured maximum at $\lambda = 5$ sits at $T_c / \omega_{\log} = 0.34$, inside the window $0.25 \leq T_c / \omega_{\ln} \leq 1.3$ they identify as anomalously shape- sensitive. We read the agreement as corroboration from an independent direction rather than as a precedent that anticipates this work.

What their work does not do, and what this paper adds, is measure a **distribution** on a constrained set. Akis and Carbotte hold one scalar index fixed and ask how far properties move as shape varies. We hold all three Allen–Dynes moments fixed, construct an explicit ensemble on the resulting moment-matched set with a declared measure, and report a standard deviation with two exact boundaries and a locatable interior maximum. Different constraint set, different quantity, same qualitative structure.

The distinction between a distribution and an extremum matters for the same reason: the extremum is a bound on what is possible, whereas a predictor’s irreducible error is governed by the spread of what actually occurs. We are likewise unaware of uncertainty quantification for T_c against a physics-computed aleatoric target; the BETE-NET dataset we use computes a bootstrapped ensemble of $\alpha^2 F$ predictions and reports its mean, without propagating the spread to T_c .

The assumption under test is stated plainly in that same literature: that “a precise knowledge of the shape of the spectral function is not really necessary, at least to a first approximation, since these quantities mostly depend on the representative frequency of $\alpha^2 F$.” We quantify how good that approximation is, and give it a floor.

A recent review of the field [Pellegrini & Sanna 2024] compares the T_c distributions produced by Eliashberg theory, density functional theory for superconductors, and the McMillan and Allen–Dynes forms, and reports that the closed forms give distributions centred on the experimental values but broader. That broadening is the quantity this paper decomposes.

(The survey above is targeted rather than exhaustive. Searches were directed at the extremal literature on the moment-matched set, at regression improvements to the closed form, at uncertainty quantification for computed T_c , and at prior measurements of spectral-shape sensitivity. We found no prior construction of an ensemble at fixed moments, which is the claim this paper rests on; a reader who knows of one should treat §4 as a re-measurement rather than a first.)

What this paper does

We solve the linearised isotropic Migdal–Eliashberg equations numerically for 806 materials from the BETE-NET DFPT database, verify the solver against ten internal checks, an analytic asymptotic limit, and an independent published Eliashberg implementation on a shared input (§3), and then:

1. measure the **learnable** component by normalising the Allen–Dynes discrepancy against the solver’s own sensitivity, which removes a parameterisation artifact that would otherwise be mistaken for a physical trend (§5), and quantify it as a cross-validated variance reduction rather than as a correlation, so that its size can be read alongside its significance (§5.6);
2. measure the **irreducible** component directly, by constructing ensembles of spectral functions matched to identical moments and computing the resulting $\sigma(\ln T_c)$ across a nine-cell grid (§4), and locate that floor between the two points at which it is zero by theory rather than by measurement (§4.1, §4.9).

What this paper compares

One framing point governs the interpretation of everything below. The reference T_c here is the Migdal–Eliashberg solution on DFT input, not the experimental T_c . Allen–Dynes is a fit to numerical solutions of those equations, so the quantity we decompose is the discrepancy between a fit and the theory it approximates, both evaluated on identical input.

This is deliberate and it is also a limitation. Our solver returns 17.1 K for Nb against a measured 9.1–9.5 K, because the BETE-NET α^2F for Nb gives $\lambda = 1.23$ where an independent DFPT calculation for the same material gives 0.94 (§3.5). Agreement with experiment is not claimed and is not the subject; a formula and the theory it approximates can be compared exactly on shared input, which is what makes the decomposition well posed.

2. Methods

2.1 The spectral function and the three moments

The electron–phonon interaction in an isotropic superconductor is fully specified, at the level of Migdal–Eliashberg theory, by a single function of frequency: the Eliashberg spectral function $\alpha^2F(\omega)$, the phonon density of states weighted at each frequency by the strength with which Fermi-surface electrons couple to those modes.

The Allen–Dynes closed form does not take α^2F . It takes three moments of it:

$$\lambda = 2 \int_0^\infty \frac{\alpha^2F(\omega)}{\omega} d\omega$$

$$\omega_{\log} = \exp \left[\frac{2}{\lambda} \int_0^\infty \frac{\alpha^2 F(\omega) \ln \omega}{\omega} d\omega \right], \quad \omega_2 = \left[\frac{2}{\lambda} \int_0^\infty \alpha^2 F(\omega) \omega d\omega \right]^{1/2}$$

All three are linear functionals of $\alpha^2 F$ (the second and third after multiplication by λ), a fact used in §4.1. Throughout we write

$$r \equiv \omega_2 / \omega_{\log}$$

for the shape parameter that enters the Allen–Dynes f_2 correction. Energies are in meV; temperatures in kelvin, with $k_B = 8.617333262 \times 10^{-2} \text{ meV K}^{-1}$.

Two invariance properties of these quantities are used later and are worth stating once. Under an amplitude rescaling $\alpha^2 F \rightarrow k \alpha^2 F$, λ scales exactly linearly while $\omega_{\log}$, ω_2 and hence r are invariant, because the latter are moments of the normalised density $g(\omega) = (2/\lambda) \alpha^2 F(\omega)/\omega$ in which k cancels. Under a pure frequency stretch $\alpha^2 F(\omega) \rightarrow \alpha^2 F(\omega/c)$, substituting $u = \omega/c$ leaves λ unchanged, while $\omega_{\log}$ and ω_2 both scale as c and r is again invariant. **λ and r therefore differ in their sensitivity to systematic error in exactly one channel: the amplitude of $\alpha^2 F$.** §3.5 measures the consequence.

2.2 The approximation under test

McMillan (1968) fitted an analytic form to his own numerical solutions of the Eliashberg equations for a scaled Nb spectrum; Allen and Dynes (1975) repeated the exercise over a wider set and added two correction factors to repair the failure of McMillan’s form at strong coupling:

$$T_c = \frac{f_1 f_2 \omega_{\log}}{1.20} \exp \left[\frac{-1.04 (1 + \lambda)}{\lambda - \mu^* (1 + 0.62\lambda)} \right]$$

$$f_1 = \left[1 + (\lambda/\Lambda_1)^{3/2} \right]^{1/3}, \quad f_2 = 1 + \frac{(r-1)\lambda^2}{\lambda^2 + \Lambda_2^2}$$

$$\Lambda_1 = 2.46 (1 + 3.8\mu^*), \quad \Lambda_2 = 1.82 (1 + 6.3\mu^*) r$$

This is an interpolation fit to numerical solutions rather than an independent theory, so what we decompose is the discrepancy between a fit and the equations it was fitted to (§1).

We use the corrected form ($f_1 f_2$ included) throughout, since it is the strongest version of the closed form, which is the appropriate baseline here. Implementation: `src/eliashberg.py:allen_dynes_tc`.

2.3 The linearised Migdal–Eliashberg equations

At $T = T_c$ the gap vanishes, so the nonlinear Eliashberg equations may be linearised in Δ . On the imaginary axis, with fermionic Matsubara frequencies $\omega_n = \pi T(2n + 1)$,

$$\lambda(n - m) = 2 \int_0^\infty \frac{\omega \alpha^2 F(\omega)}{\omega^2 + (\omega_n - \omega_m)^2} d\omega$$

$$Z_n = 1 + \frac{\pi T}{\omega_n} \sum_m \lambda(n - m) \text{sgn}(\omega_m)$$

$$Z_n \Delta_n = \pi T \sum_m \left[\lambda(n - m) - \mu^* \right] \frac{\Delta_m}{|\omega_m|}$$

The last line is a linear homogeneous eigenproblem $\Delta = \mathcal{K}(T)\Delta$ with a nontrivial solution only where $\mathcal{K}(T)$ has unit eigenvalue. Writing $\rho(T)$ for the largest eigenvalue of $\mathcal{K}(T)$, T_c is defined by $\rho(T_c) = 1$, with $\rho > 1$ the superconducting side. We bracket and bisect on T to a relative tolerance of 2×10^{-3} .

2.4 Numerical implementation

Three implementation choices are load-bearing and are stated because the verification in §3 tests them individually.

Toeplitz structure. $\lambda(n - m)$ depends only on the index difference, so $\mathcal{K}$ is a symmetric-Toeplitz operator plus a rank-one term. Embedding the first column in a circulant of length $\geq 2N - 1$ turns the matrix–vector product into a circular convolution evaluated by real FFT, giving $\mathcal{O}(N \log N)$ time and $\mathcal{O}(N)$ memory against $\mathcal{O}(N^2)$ for both in the dense form.

The μ^* term is rank one. Because μ^* is constant in (n, m) , its contribution to $\mathcal{K}$ is $-\mu^* (1/Z_n) \otimes (1/|\omega_m|)$, whose action on a vector is a scaled sum. It is therefore applied analytically rather than through the convolution.

Largest real part, not largest magnitude. T_c is set by the largest *real* part of the spectrum of $\mathcal{K}$, whereas power iteration converges to the largest *magnitude*. The μ^* rank-one term is negative and its magnitude grows with μ^* ; above $\mu^* \approx 0.25$ (measured; see T2d in §3.1) the resulting negative eigenvalue overtakes the positive one, and power iteration then converges cleanly — without stalling, so an iteration-count guard does not fire — onto a value that a naive reading interprets as “not superconducting.” We therefore fall back to Arnoldi with `which="LR"` whenever the power iteration stalls **or** returns $\rho \leq 0$. The second condition is logically complete: a negative value cannot be the largest real part while any positive eigenvalue exists, and where none exists the Arnoldi call returns the correct negative value anyway.

The number of Matsubara frequencies scales as ω_c/T , so the cost of the matrix-free formulation is what makes an uncapped sum affordable. We set the backstop at $N_{\max} = 2.5 \times 10^5$, above the requirement of the most demanding material in the set ($n_{\text{cut}} = 22,380$ for BeGeSc at $\omega_c = 10 \omega_{\max}$); no material in the reported dataset is capped. A separate numerical floor $T_{\text{floor}} = 5 \times 10^{-3}$ K exists only to stop the bracket search walking to zero and is **not** a criterion for superconductivity; the reporting threshold is a distinct, downstream constant (§2.6) which we vary and report the effect of.

Implementation: `src/eliashberg.py`.

2.5 The Coulomb pseudopotential and its cutoff

μ^* has no meaning without the energy cutoff at which it is defined. The Morel–Anderson result

$$\mu^* = \frac{\mu}{1 + \mu \ln(E_F/\omega_c)}$$

makes μ^* an increasing function of ω_c , so quoting “ $\mu^* = 0.1$ ” without stating ω_c is incomplete. Allen and Dynes fitted their formula against solutions using their own convention; we solve with the Matsubara sum carried to $\omega_c = 10 \omega_{\max}$. Because T_c depends on μ^* through the exponential denominator $\lambda - \mu^*(1 + 0.62\lambda)$, a small convention mismatch is amplified, and amplified most at small λ where the μ^* term is a larger fraction of the denominator.

We therefore calibrate rather than assume. For a set of Einstein spectra we solve for the μ_{ME}^* at our cutoff that reproduces the Allen–Dynes T_c evaluated at μ_{AD}^* , and read the result off the plateau $\lambda \leq 0.8$:

$$\mu_{\text{AD}}^* = 0.10 \longleftrightarrow \mu_{\text{ME}}^* = 0.1293, \quad \mu_{\text{AD}}^* = 0.13 \longleftrightarrow \mu_{\text{ME}}^* = 0.1840$$

the second pair supplying the finite difference for the sensitivity derivative of §5.2. Einstein spectra are used deliberately: a single sharp mode is the case closest to what the closed form was fitted against, so residual

drift cannot be blamed on exotic spectral shape. §3.3 validates this calibration against an independently published analytic relation.

Implementation: `src/diagnose_mustar.py`.

2.6 Dataset

Input spectra are the $\alpha^2 F(\omega)$ of the BETE-NET database (Hennig group), computed by DFPT. Gibson *et al.* report $\alpha^2 F$ for 818 dynamically stable materials; the released database contains 806, all of which have finite positive λ and all of which we use. For each we compute the three moments from the tabulated $\alpha^2 F$ by trapezoidal quadrature (verified against the database’s own stored moments, §3.1), the Allen–Dynes T_c , and the numerical Migdal–Eliashberg T_c at both μ_{ME}^* values of §2.5.

No material is Matsubara-capped. Of the 806, 520 have T_c^{ME} above the reporting threshold $T_{\text{rep}} = 0.05$ K and 583 have a defined $\ln(T_c^{\text{ME}}/T_c^{\text{AD}})$. T_{rep} is applied downstream of the solver and is swept; §5.4 shows that it moves both the magnitude **and the sign** of the mean Allen–Dynes bias, so all reported error statistics name their cut.

Implementation: `src/build_physics_dataset.py`.

3. Verification

A solver written for a single paper by a single author invites, reasonably, the question of whether it is right. We answer it in four ways: internal checks against independent sources of truth (§3.1), an analytic limit with no fitted parameters (§3.2), an external benchmark against an independently published Eliashberg implementation on a shared input (§3.4), and an external check of the μ^* convention against a published analytic relation (§3.3).

3.1 Internal checks

Each check has a source of truth outside the code being tested. All ten pass. Reproduced independently on a second machine and a different Python, numpy and scipy stack; agreement to the last reported digit except where noted.

	test	source of truth	result
T1	moment quadrature	database’s stored $\lambda, \omega_{\log}, \omega_2$	max rel. dev. 1.6×10^{-3} over 200 materials
T2	power iteration	exact dense eigendecomposition	1.5×10^{-11}
T2b	matrix-free matvec	dense matvec, random vectors	5.5×10^{-15}
T2c	Arnoldi fallback	exact dense eigendecomposition	1.9×10^{-15}
T2d	largest real part	exact spectrum, μ^* swept 0.10→0.32	4.6×10^{-11} ; negative-dominant in 5/12 cases, onset $\mu^* = 0.25$
T3	Matsubara cutoff, matched pairs	self-consistency under Morel–Anderson rescaling	matched increment 0.098% (Nb), 0.157% (Al) at 20→40; fixed- μ^* increments constant to 4.3% and 1.4%

	test	source of truth	result
T4	strong-coupling asymptote	Allen & Dynes 1975 eq. 5.7, analytic	$T_c/T_c^{\text{asym}} = 1.0022$ at $\lambda = 100$
T5	monotonicity of $\rho(T)$	— (well-posedness of the bisection)	pass
T6	weak coupling vs Allen–Dynes	the fit’s own domain of validity	median $T_c^{\text{ME}}/T_c^{\text{AD}} = 0.985$, $n = 40$
T7	μ^* cutoff rescaling	Morel–Anderson logarithmic relation	0.59%

T2b deserves comment because it is the check that would catch what T2 cannot. T2 compares eigenvalues, which integrate over the whole operator; an indexing error in the Toeplitz embedding can perturb the spectrum by less than T2’s threshold and survive. T2b compares the operator itself applied to random vectors, at three Matsubara counts and three values of μ^* , with the resulting matrix sizes asserted so that a test silently running at the wrong N cannot pass.

T5 as implemented tests one material at one μ^* . We additionally verified monotonicity across 25 randomly drawn materials at $\mu^* \in \{0.10, 0.1293, 0.1840\}$, 20 temperatures each spanning $0.2\text{--}2.5 T_c^{\text{AD}}$: no violations.

T6 is a transfer test, not a tautology — the constant of §2.5 is read off Einstein spectra and applied here to real materials — but its tolerance is wider than it appears. The median ratio moves with μ_{ME}^* at $\partial(\text{median})/\partial\mu^* \approx -6.0$, so the assertion $|\text{median} - 1| < 0.10$ admits $\mu_{\text{ME}}^* \in [0.113, 0.146]$. It confirms the transfer to ± 0.017 in μ^* , and it constrains only the median — it carries no information about drift with λ or r .

3.2 The strong-coupling limit

As $\lambda \rightarrow \infty$ the Eliashberg equations admit an analytic asymptotic solution, $k_B T_c \rightarrow 0.1827 \sqrt{\lambda \langle \omega^2 \rangle}$, derived independently of any fit and containing no adjustable parameters. It is the strongest available check because McMillan’s exponential form *saturates* in this limit while the true T_c grows without bound: a solver that reproduced the fit rather than the equations would fail here.

At $\mu^* = 0$, Einstein spectra, $\omega_c = 20 \omega_{\text{max}}$:

λ	5	10	25	50	100
$T_c^{\text{ME}}/T_c^{\text{asym}}$	0.9351	0.9715	0.9917	0.9986	1.0022

Monotone approach to unity, 0.2% at $\lambda = 100$.

3.3 The μ^* convention against a published analytic relation

Kogler *et al.* (IsoME, 2025) derive the cutoff dependence of the pseudopotential as $1/\mu_{\text{E}}^* = 1/\mu_{\text{AD}}^* + \ln(\omega_{\text{ph}}/\omega_c)$, with ω_{ph} the highest phonon frequency. With our $\omega_c = 10 \omega_{\text{max}}$ the ratio is 1/10 for every material, so the relation predicts a single universal constant — the same structure our calibration has.

Agreement at one value of μ_{AD}^* would be weak evidence, since any calibration landing near 0.13 would match. We therefore recalibrated from scratch at five values and compared the functional form:

μ_{AD}^*	0.080	0.100	0.110	0.130	0.150
analytic μ_{E}^*	0.09806	0.12991	0.14731	0.18554	0.22914

μ_{AD}^*	0.080	0.100	0.110	0.130	0.150
this work	0.09823	0.12928	0.14608	0.18347	0.22725
deviation	+0.17%	−0.49%	−0.84%	−1.11%	−0.83%

Agreement is better than 1.2% across a factor 2.3 in μ_{E}^* . Inverting for the implied ratio gives $\omega_{\text{ph}}/\omega_c = 0.104$ against a nominal 0.100; the 4% residual maps to the $\sim 0.9\%$ seen in the table and we do not distinguish between leading-logarithm error in the Morel–Anderson relation and a small offset in our own calibration.

One implementation detail matters for the comparison: the IsoME code takes ω_{ph} as the highest frequency at which $\alpha^2 F$ exceeds a threshold of 0.01 rather than the grid maximum; on their own Nb spectrum these differ by a factor 0.82. Our measured ratio, 0.104, is closer to the unthresholded reading (nominal 0.100) than to the thresholded one (0.091).

3.4 External benchmark

The checks above are internal or analytic. To test the solver against an independent numerical implementation we use IsoME, an open-source Julia Eliashberg package whose repository ships the $\alpha^2 F$ used for its published Nb result.

Benchmarking requires holding the input fixed, which is the reason for using their spectrum rather than ours: their T_c^{AD} and ours are the *same closed form*, so any disagreement on that row is attributable to input alone. On their Nb file (smearing column 15, within the converged plateau of λ ; see §3.5) we obtain $\lambda = 0.9377$, $\omega_{\text{log}} = 13.504$ meV, $\omega_2 = 15.942$ meV, $r = 1.1806$.

We then run our solver at *their* conventions with no translation: constant density of states, their Matsubara cutoff $\omega_c = 7000$ meV, and μ^* obtained from their own conversion.

	this work	IsoME
$T_c^{\text{AD}}, \mu_{\text{AD}}^* = 0.12$	9.30 K	9 K
$T_c^{\text{ME}}, \mu^* = 0.360$ (their conversion)	9.10 K	9 K
$T_c^{\text{ME}}, \mu^* = 0.38$ (their GW value)	8.99 K	9 K
experiment		9.1–9.5 K

All three fall inside their published integer, and the result is insensitive to which branch of their μ logic is taken.

Three limitations must travel with this table. **(i)** Their T_c is tabulated to integer kelvin, so at 9 K the comparison has a resolution of $\pm 5.6\%$ — one to two orders of magnitude looser than the internal checks of §3.1–3.3. Its value is that it is external, not that it is precise. **(ii)** It exercises the solver at $\omega_c/\omega_{\text{max}} \approx 212$, not the production value of 10; §3.6 supplies the cross-cutoff consistency that connects them. **(iii)** The T_c^{AD} row is a check that we parse their 2000×20 tabulation and select the smearing column correctly — valuable, but a parse check, not agreement on physics. Only the two T_c^{ME} rows test the solver.

3.5 Sensitivity of the moments to DFT convergence

The same file supplies a controlled measurement of something needed in §6. IsoME’s Nb tabulation carries 20 double-delta smearing widths for one material computed with one code, one k -mesh and one pseudopotential, so the smearing sweep isolates a single axis of DFT convergence error with everything else held fixed.

	λ	$\omega_{\log}$	ω_2	r
relative spread, all 20 columns	44.0%	2.05%	1.50%	0.59%
relative spread, converged plateau (cols 13–17)	0.80%	0.09%	0.03%	0.06%

λ is $75\times$ more sensitive than r across the full sweep and $13\times$ within the converged plateau. The mechanism is the amplitude invariance of §2.1: $\text{corr}(\lambda, \max \alpha^2 F) = 0.9989$ across the sweep, and dividing the amplitude out reduces λ ’s spread from 44.0% to 7.0%.

Consistently, the two DFT pipelines disagree far more on λ than on r for the same material:

Nb	λ	$\omega_{\log}$	ω_2	r
IsoME (QE/EPW)	0.9377	13.504	15.942	1.1806
BETE-NET	1.2325	14.553	16.822	1.1559
difference	+31.4%	+7.8%	+5.5%	−2.1%

We stress that this is a single material and a single pair of pipelines, and we do not adjudicate which is correct. The claim is only that the two disagree by 31% on λ and 2% on r , consistent in direction and order with the controlled smearing result.

3.6 What is, and is not, convergent in the cutoff

At *fixed* μ^* , T_c cannot converge as $\omega_c \rightarrow \infty$. The Coulomb kernel is constant in (n, m) , so its Matsubara sum grows as $\ln(\omega_c/T)$ while the phonon kernel decays and its sum converges. Measured on Nb at $\mu^* = 0.10$, the increment in T_c per doubling of ω_c is constant to bisection resolution from $\omega_c/\omega_{\max} = 10$ to 160 (+0.25 K per doubling), i.e. $T_c \simeq \text{const} + 0.36 \ln \omega_c$. Setting $\mu^* = 0$ removes the effect entirely (steps fall to −0.018%), confirming the mechanism.

Only the *pair* $(\omega_c, \mu^*(\omega_c))$ is physical. Rescaling μ^* between cutoffs by the Morel–Anderson relation and comparing matched pairs, the solver is consistent:

$\omega_c/\omega_{\max}$	10	20	40
μ^*	0.1293	0.14203	0.15754
Nb, matched	—	−0.46%	−0.56%
Nb, fixed μ^*	—	+2.25%	+4.59%
Al, matched	—	−0.76%	−0.91%
Al, fixed μ^*	—	+9.67%	+19.20%

The matched increments shrink; the fixed- μ^* increments are constant per doubling. Al is the extreme case because low λ amplifies μ^* . Over 36 materials the matched-pair deviation between $\omega_c/\omega_{\max} = 10$ and 20 has median −0.40% and worst case −0.89%.

T3 in the verification suite asserts both legs of this: that the matched increment falls below 0.2% between $\omega_c/\omega_{\max} = 20$ and 40, and that the fixed- μ^* increment is constant per doubling to within 15%. A test that asserted only convergence would pass on a threshold’s placement; asserting the non-convergence too makes the failure mode diagnostic.

Implementation: `src/matched_cutoff.py`.

3.7 What the solver does not claim

It solves the isotropic Migdal–Eliashberg equations at constant electronic density of states on the imaginary axis, with a constant μ^* . It does not treat anisotropy, a variable density of states, vertex corrections, or the frequency dependence of the screened Coulomb interaction. Its output is the T_c implied by the DFT input it is given. For Nb that input yields 17.1 K at production settings against 9.1–9.5 K measured, the λ responsible being 31% above an independent pipeline’s for the same material (§3.5).

4. The irreducible component

Allen–Dynes claims that three moments of α^2F determine T_c . They do not. Infinitely many spectral functions share the same $(\lambda, \omega_{\log}, \omega_2)$ and have different true T_c , and that spread is a hard limit on any predictor taking only those three inputs. This section measures it.

4.1 Construction

The three moments are linear functionals of α^2F (§2.1), so the moment-matched set $\{\alpha^2F \geq 0 : A \alpha^2F = b\}$ is a convex polytope. We verified this operationally: walking the segment between two moment-matched spectra leaves all three moments matched to 10^{-16} at every step.

Convexity gives a well-posed *set*. It does not give a measure on that set, and a spread requires one. This is the central methodological difficulty of the section and we return to it in §4.6.

The set also has an exact boundary that fixes one endpoint of everything below. For any positive measure, Jensen’s inequality gives $\omega_{\log} = \exp(\ln \omega) \leq \langle \omega \rangle$ and the power-mean inequality gives $\langle \omega \rangle \leq \langle \omega^2 \rangle^{1/2} = \omega_2$, so

$$r \geq 1, \quad \text{with equality iff all spectral weight sits at a single frequency.}$$

At $r = 1$ the moment-matched set is therefore a **single point** — the Einstein spectrum — and the irreducible spread is exactly zero, by inequality rather than by measurement. (Empirically, $\min r = 1.026$ over the 806 spectra, with no violations.) The r -dependence reported in §4.4 is consequently not a bare empirical correlation: it is the interior of a curve with a theoretically fixed endpoint at $r = 1$, and the classical result that the Einstein spectrum maximises T_c at fixed λ and $\langle \omega^2 \rangle$ is that endpoint stated in extremal language.

The measure we adopt is empirical. Writing each spectrum in scale-free form $\tilde{g}(u)$ with $u = \omega/\omega_{\log}$, all shape information at fixed moments is carried by $\tilde{g}$ at fixed r , and the 806 BETE-NET spectra are 806 draws from $p(\tilde{g} | r)$. Each donor is moved onto the target moments by exponential tilting,

$$\tilde{g} \mapsto \tilde{g} \cdot u^b e^{cu^2}$$

which is the I -projection: among densities satisfying the constraints, the one of minimum Kullback–Leibler divergence from the donor. It is therefore the smallest deformation consistent with the target, and non-negative by construction. Achieved moment errors are 10^{-11} – 10^{-16} . No donor in 7,254 was rejected for moment error.

4.2 Reachability is one-sided, and this sets a hard ceiling

Exponential tilting is absolutely continuous with respect to the donor: it reweights existing support and cannot create support. A donor carrying no spectral weight at high frequency therefore cannot be tilted to a high r at any tilt strength.

The consequence is directional, and measuring it rather than assuming it changes what it implies. Failure is **not** symmetric in $|\ln(r_{\text{donor}}/r_{\text{target}})|$; it is one-sided in the *sign*. Across all nine targets, donors with r below the target fail almost always and donors with r above the target almost never fail, at any distance:

target	r	fail rate, donors below target	fail rate, donors above target
CrRh3	1.057	78%	0–8%
AlHfNi2	1.303	97–100%	0–1%
Se2V	1.561	96–100% (below -0.20)	0%

Tilting can lower r but not raise it, with about 10–13% of headroom. The reachable set at a target is therefore, to good approximation, $\{\text{donors with } r \gtrsim 0.88 r_{\text{target}}\}$, which is computable from the donor r -distribution without solving anything. A reachability map built on that rule predicted the achieved donor count at all nine targets exactly (e.g. CrRh3 763, BeNi2Si 366, Be2PdZn 254).

Because donor r is right-skewed across the 806 (median 1.223, 90th percentile 1.538), the reachable count falls monotonically with target r : $\rho(r, n_{\text{reach}}) = -1.000$ exactly across the nine cells. **Sample size is perfectly anti-correlated with the axis carrying the signal**, which is why every result below is reported with an explicit control for it.

4.3 The grid

Nine targets on a 3×3 tertile grid in (λ, r) , each evaluated against **every** one of the 806 donors — exhaustive, not sampled, so there is no donor-selection variance to disentangle. $\sigma(\ln T_c)$ is taken over the reachable set at each target.

target	r	λ	n	$\sigma(\ln T_c)$	κ	precision
CrRh3	1.057	0.469	763	0.00202	10.44	6.4%
HfRu2Ta	1.070	0.742	752	0.00105	26.51	9.7%
CoTi	1.161	1.012	601	0.00315	2.73	4.4%
AsZr	1.213	0.859	501	0.00320	7.91	7.0%
AlHfNi2	1.303	0.593	397	0.00955	6.09	7.1%
BeNi2Si	1.333	0.411	366	0.01844	3.39	6.1%
Ga2LiRu	1.410	0.744	306	0.00847	6.04	8.1%
Be2PdZn	1.483	0.444	254	0.02421	2.78	7.6%
Se2V	1.561	1.250	207	0.01987	2.98	7.8%

The precision column is the relative standard error of $\hat{\sigma}$, $\sqrt{(2 + \kappa)/4n}$. It is set by κ at least as much as by n : the distribution of $\ln T_c$ over moment-matched spectra is leptokurtic at every cell, so $\hat{\sigma}$ is a second moment estimated on a heavy tail. **$\pm 10\%$ is achieved at all nine cells; $\pm 5\%$ at one.** The two cells that cannot reach $\pm 5\%$ are the two with the *most* donors — CrRh3 with 763, 92% of the entire pool — because $\kappa = 10.4$ and 26.5 there demand $n \approx 1200$ and 2900. More donors would not fix the cells that need fixing.

$\hat{\kappa}$ is itself the least precisely determined quantity here: its normal-theory standard error is $\pm 18\%$ at $n \approx 750$ and worse under the tails actually present. Sample sizes derived from it should be read to one significant figure.

4.4 The floor rises with the spectral shape parameter

$$\rho(\sigma, r) = +0.917 \quad (p = 0.0005), \quad \rho(\sigma, r \mid \lambda) = +0.951 \quad (p = 0.0003)$$

The design decorrelates the two covariates — $\rho(\lambda, r) = +0.083$ across the nine cells — so the partial and the marginal agree. The relation is strong but not monotone: four of 36 ordered pairs invert.

Three controls, each of which the result survives.

Leave-one-out. +0.917 to +0.970 across all nine deletions, $p \leq 0.0037$ everywhere. No single cell carries it.

Matched sample size. Because $\rho(r, n) = -1.000$, “the floor rises with r ” and “the floor rises as n falls” are the same statement on this grid. Subsampling every cell’s $\ln T_c$ list to a common $n = 207$ over 4,000 draws and recomputing: $\rho(\sigma, r) = +0.917$, unchanged to three decimals, with the largest shift in any cell’s $\hat{\sigma}$ being 1.1% and the cell ordering preserved. The small-sample bias of a second moment is $O(1/n)$ in relative terms and is simply too small to generate a 23-fold spread.

Matched donor composition. Restricting all nine targets to the 193 donors reachable at every target, so that only the moments vary: $\rho(\sigma, r) = +0.883$ ($p = 0.0016$), $\rho(\sigma, r \mid \lambda) = +0.943$, leave-one-out +0.901 to +0.952 with $p \leq 0.0056$ throughout.

The trend is robust to all three. This is the section’s result.

Functional form. A trend reported as linear in r would contradict the boundary condition of §4.1, since extrapolating it to $r = 1$ gives a nonzero floor where the inequality requires exactly zero. We therefore fit the minimal form that respects the boundary,

$$\sigma \sim (r - 1)^p e^{b\lambda},$$

which on the nine cells gives $p = 1.31$, $b = -1.01$, $R^2 = 0.902$ — a better fit than the linear-in- r alternative ($R^2 = 0.919$ with three free parameters against two, and violating the boundary). §4.8 compares these exponents against the same form fitted to real materials.

4.5 The coupling constant is not resolved on this grid

$\rho(\sigma, \lambda) = -0.183$ marginally and $\rho(\sigma, \lambda \mid r) = -0.652$ ($p = 0.080$) controlling for shape. The partial is suggestive and we do not claim it. Leave-one-out is decisive against claiming it: deleting Be2PdZn alone takes the partial to -0.398 at $p = 0.377$, while every other deletion leaves it between -0.605 and -0.723 . A coefficient that one deletion of nine undoes is not a result.

We note this explicitly because an earlier four-cell subset of this same grid gave $\rho(\sigma, \lambda \mid r) = +0.800$ — the opposite sign — for a mechanical reason: within those four cells $\rho(\lambda, r) = +0.800$ against +0.083 on the full grid, so λ and r were collinear and no attribution between them was possible. Resolving λ would require cells this grid does not contain; more donors in the cells it does contain will not help. §4.9 builds those cells synthetically, holding r fixed by construction instead of partialling it out, and does resolve it — with the opposite sign to the cross-cell fit.

4.6 The magnitude is measure-dependent, and that is the finding

The spread across the nine cells is $23.1\times$ under the natural donor set at each target and $14.1\times$ under the common 193. Both are correct; they estimate different quantities.

The common set is not a neutral control. To be reachable at every target a donor must satisfy $r \gtrsim 1.37$, so the common set is the top quarter of the pool by r (median donor $r = 1.478$ against 1.223 for the pool). The effect of restricting to it is therefore one-sided: $\hat{\sigma}$ rises by 44% and 65% at the two lowest- r targets and by 0–7% at the four highest, because for the high- r targets the common set already *is* the natural set. Restricting to it forces the low- r cells onto donors far above their target — precisely the large deformation whose effect the comparison is meant to isolate.

Nor does reweighting escape the restriction. Matching each target’s reachable set to a common donor- r density by importance weights fails because at high r the low- r donors are *absent* rather than merely rare: any common target density must live inside the overlap band, which returns the common set with added variance. **The binding constraint is support, not weighting.**

The two magnitudes are therefore two declared measures — and, more usefully, they have different jobs:

- $23.1\times$ and its per-cell components — the floor **at a given material**, computed over the donors that can actually reach that material’s moments. This is the quantity that enters any per-cell comparison against a predictor’s error, because the question there is what the floor is at that material.
- $14.1\times$ — the **cross-cell trend**, computed at one fixed donor composition. This is the version in which r is not confounded with which donors were used, and is therefore the one that supports §4.4.

Assigning each number to the job it is valid for is stronger than reporting both as alternatives. What neither supports is a single measure-free magnitude: convexity gives a well-posed set but no canonical measure on it, and the choice of donor population *is* a choice of measure. The gap between the two figures — roughly 40% of the dynamic range — measures how much the answer depends on that choice, and we report it as such.

4.7 Comparing the floor with the Allen–Dynes error

The ratio of the Allen–Dynes error to this floor is quoted in §6.3 with a table of how it moves under each alternative declaration. Three choices govern it, and their sizes are measured rather than asserted:

choice	what it is	worth
recalibration class	how rich a function of the same three moments the numerator may be. Its trivial member is “standard deviation, bias reported separately”, which is forced: the floor is a spread about each cell’s own centre and carries no mean term, whereas $\text{RMS}^2 = \bar{\varepsilon}^2 + \text{sd}^2(n-1)/n$ imports the bias (§5.4). Refitting an intercept sets the residual mean to zero, so the intercept-only residual sd <i>is</i> $\text{sd}(\varepsilon)$, to all ten digits we carry	$1.82\times$ (§5.6)
population	the cut defining a superconductor	$2.65\times$ on the numerator; on the denominator a signed coverage effect of $0.42\text{--}0.99\times$ (§6.3)
convention	the value of μ^*	$1.59\text{--}3.26\times$ (§5.4)

The population choice is not symmetric between the two sides. On the numerator it is a genuine choice. On the denominator it is only a coverage question, because σ is a function of a moment triple rather than of a population: every cell is measured by tilting all 806 donors onto its target moments, so the cut a target was drawn from does not enter its value. §6.3 measures the resulting coverage gap.

One detail of the dataset build is worth recording, since it amounts to a cost bound applied to the arm under test. The generator short-circuits the solver when the closed-form estimate is small, `tc_me = 0.0` if `tc_ad < SOLVER_FLOOR_K` else `eliashberg_tc(...)`, skipping 223 of 806 materials. Solving the skipped set directly returns exactly zero for 98 of them and a value between 0.0057 and 0.0091 K for five; none reaches the 0.05 K reporting threshold, so no number here changes. We have left it in place rather than patched it, since the population precondition above supersedes it in any case.

4.8 The donor construction is a lower bound; real materials give the upper

Everything above measures the spread over a *constructed* ensemble. The quantity a predictor actually faces is the spread over the **physical** conditional distribution of $\alpha^2 F$ given the moments, and the two need not coincide. Because exponential tilting reweights support and cannot create it (§4.2), the constructed ensemble is necessarily the narrower of the two: §4.3’s numbers are a **lower bound**.

The physical spread can be measured without any construction. Allen–Dynes is a function of the moments alone, so for two materials with identical moments $\Delta\varepsilon = \Delta \ln T_c^{\text{ME}}$ exactly. Pairs of real materials near-matched in all three moments therefore expose the conditional spread with the moment difference cancelled to first order. Over the same $T_c^{\text{AD}} > 1$ K population as §4.3 ($n = 304$), matching each moment to a relative tolerance τ and estimating $\hat{\sigma} = \text{sd}(\Delta\varepsilon)/\sqrt{2}$:

τ	pairs	$\hat{\sigma}$ from $\Delta\varepsilon$	$\hat{\sigma}$ from $\Delta \ln T_c^{\text{ME}}$
3%	20	0.0172	0.0504
5%	68	0.0301	0.0796
10%	362	0.0278	0.1478
20%	1909	0.0295	0.2793

The two columns are the argument. $\Delta\varepsilon$ is **flat** across a sevenfold range of tolerance, while the uncontrolled $\Delta \ln T_c^{\text{ME}}$ over the same pairs grows by a factor of 5.5 — as it must, since looser matching admits larger genuine moment differences. A quantity that does not grow as the matching is loosened is not leaked moment difference; it is conditional spread.

Comparing with the nine-cell donor floor (0.00105–0.02421, median 0.00847), the construction understates the physical conditional spread by roughly a factor of three, in the direction the support argument predicts. Taking the constructed value as a lower bound and the pair estimate as an upper bound — the latter inflated because the tabulated moments carry their own error, which enters $\Delta\varepsilon$ as noise — gives

$$0.0085 \lesssim \sigma_{\text{irreducible}} \lesssim 0.028.$$

Both ends are computed at the same solver tolerance. This is not automatic: the donor ensembles were solved at a bisection tolerance of 10^{-4} while the material dataset was built at the signature default of 2×10^{-3} , twenty times looser. At the looser setting T_c^{ME} is visibly quantised — 92 distinct values of ε among 304 materials, with 91% of rows sharing a value — and pairs of *unrelated* materials acquire identical ε by landing in the same bisection bin. Rebuilding the population at 10^{-4} (29 s; median T_c shift 3.8×10^{-4}) raises the distinct count to 272 and reduces tied rows to 20%. The bracket above is quoted from the rebuilt values; it is unchanged to three decimals at every matching tolerance, because a variance is dominated by pair differences far above the grid step and the ties sit in a tail that contributes almost nothing to it. What the ties *do* affect is discussed below.

Both §4.4 trends reproduce on this construction-free data. Splitting the matched pairs by shape, $\hat{\sigma} = 0.0212$ for $r < 1.2$ against 0.0349 for $r \geq 1.3$ — a factor 1.64, the same sign and comparable magnitude to the +0.92 of §4.4. Within a narrow shape band the coupling constant enters negatively, at a ratio close to the cleanest matched pair of §4.5. **The two correlations of §4.4 and §4.5 are therefore not artifacts of the donor construction**, since they appear in real materials whose moments were never manipulated.

Fitting the boundary-respecting form of §4.4 to both ensembles separates the shape exponent cleanly and the coupling coefficient not at all:

	p (exponent in $r - 1$)	b (coefficient of λ)
real near-matched pairs	0.61 – 0.80	not quoted
donor construction, nine cells	1.31	–1.01

The exponent looks like a property of the data. The coupling coefficient looks like a property of our analysis choices, and we do not report a value for it. §4.9 makes the case against it stronger still: measured at fixed r instead of fitted across cells, the coupling dependence comes out with the opposite sign.

The exponent is stable: 0.61–0.80 across matching tolerances from 5% to 20% and across least-squares and median fits, with the construction’s 1.31 outside a material-clustered bootstrap interval at every setting. That is a resolved difference between the constructed and physical ensembles: **the construction is steeper in r than the materials it stands in for.**

The coupling coefficient is not. It drifts monotonically by a factor 2.5 with the matching tolerance — a parameter fixed by convention rather than by argument — and before the tolerance rebuild described above it drifted by a further 1.4 between least-squares and median fits, because $\ln |\Delta\varepsilon|$ is unbounded below and the bisection ties placed unrelated materials at $\Delta\varepsilon = 0$. The rebuild removes the second sensitivity (least-squares and median now agree to 3% at 10% matching) but not the first. A range whose endpoints are set by an unargued convention is a measurement of that convention, not of λ , so we report no value. Whether the construction is also shallower in λ than the physical ensemble is open, and §6.5 describes the experiment that would settle it.

4.9 The floor vanishes at both ends of the shape–coupling plane

Section 4.1 established one exact boundary: $\sigma = 0$ at $r = 1$, where the moment-matched set collapses to a point. There is a second, at the opposite corner, and it follows from the same asymptotic result that anchors the solver in §3.2.

As $\lambda \rightarrow \infty$ the Eliashberg equations give $k_B T_c \rightarrow 0.1827 \sqrt{\lambda \langle \omega^2 \rangle}$. Both arguments of that limit — λ and $\langle \omega^2 \rangle = \omega_2^2$ — are moments held fixed by the matching. **Every moment-matched spectrum therefore converges to the same T_c , and $\sigma \rightarrow 0$ as $\lambda \rightarrow \infty$.** The floor is therefore not monotone in λ : whatever it does at accessible coupling, it must return to zero at strong coupling. The argument says nothing about the low- λ end. In particular it gives no reason to expect σ to vanish there, and the measurement below shows that it does not.

Real materials cannot test this — only 10 of the 806 have $\lambda \geq 1.5$ — but synthetic targets can. At $r = 1.19$ held **exactly**, exhaustively over all 552 reachable donors at each rung:

λ	0.40	0.50	0.60	0.70	0.90	1.20	1.60	2.12	3.00	5.00	10.0	20.0
$\sigma(\ln T_c)$	0.01055	0.00765	0.00559	0.00407	0.002820	0.00588	0.01059	0.01440	0.01715	0.017830	0.01527	0.01140

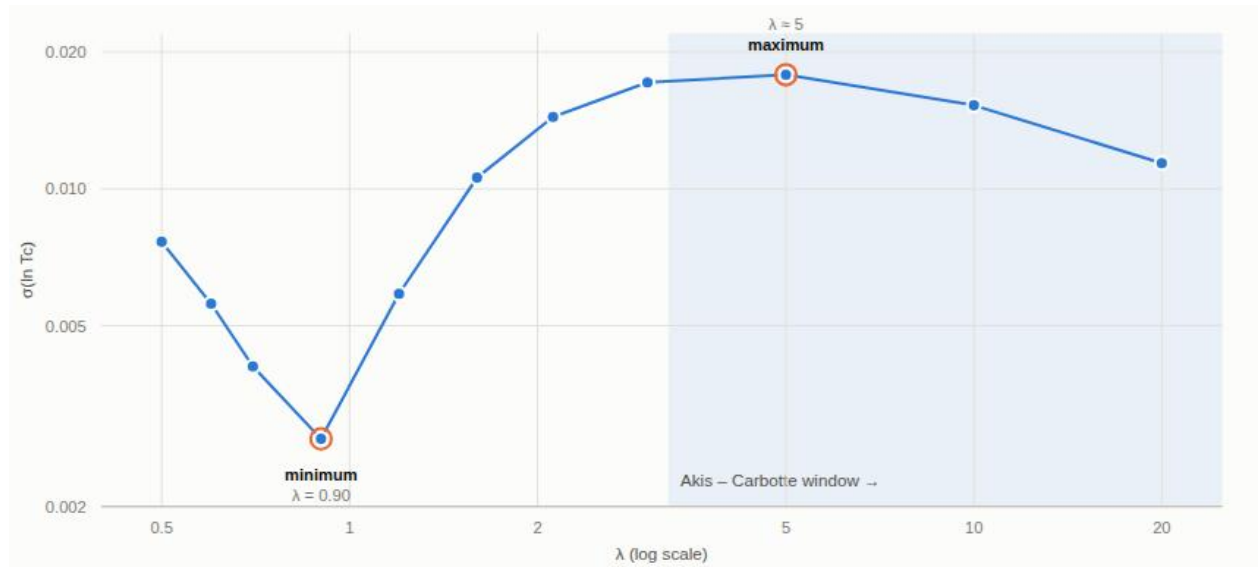


Figure 1: $\sigma(\ln T_c)$ over moment-matched spectra at $r = 1.19$ held exactly, 552 donors per rung.

There are two turning points here rather than one: a minimum at $\lambda \approx 0.90$ and a maximum at $\lambda \approx 5$. Below the minimum σ rises as coupling falls, with a log-slope of -2.67 over the five synthetic points from 0.40 to 0.90. This is the first measurement of the coupling dependence in the paper in which r is held fixed by construction instead of being partialled out, and both turning points come from synthetic targets alone.

Our maximum at $\lambda = 5$ converts to $T_c/\omega_{\log} = 0.34$, inside the window Akis and Carbotte flag as anomalously shape-sensitive (§1).

The low- λ rise looks like the amplification of §5.1 turning up again in the irreducible term: as λ falls the exponential denominator of the transition temperature collapses, so a fixed dispersion in spectral shape at fixed moments produces a growing dispersion in $\ln T_c$. We do not claim a low- λ boundary. The strong-coupling zero is an asymptotic result; the weak-coupling end is measured, and what is measured is a rise.

This settles the status of the nine-cell coefficient. §4.8’s cross-cell fit gives $b = -1.01$, σ falling with λ over the whole accessible range. Across the ladder’s range the sign is the opposite of that above the minimum and agrees only by accident below it. The cause is confounding: across the nine cells λ and r co-vary, so a coefficient fitted there was never $\partial \ln \sigma / \partial \lambda$ at fixed r . §4.8’s refusal to report b was right for a weaker reason than the real one — it is not merely convention-dependent in magnitude, it is wrong in sign.

A second construction provides a cross-check. An independent ladder built on *real-material* targets rather than synthetic triples spans $\lambda = 0.482\text{--}0.832$ at $r \in [1.170, 1.209]$ — near 1.19 but not at it, which is precisely the confound just diagnosed. Correcting each point to $r = 1.19$ through §4.8’s fitted exponent changes it by at most 15% and makes the descending limb *steeper* ($-2.59 \rightarrow -2.99$). At $\lambda = 0.60$ the corrected real-material value is 0.00558 against 0.00559 for the synthetic rung at exactly $r = 1.19$: two independent target constructions, one carrying a fitted r correction and one needing none, agreeing to 0.2%. That validates the r exponent, the synthetic-target construction and the ladder simultaneously.

Control. Every comparison across rungs assumes σ is invariant to the overall frequency scale, which §2.1 asserts and we had not measured. Holding $\lambda = 2.118$ and $r = 1.19$ fixed and varying $\omega_{\log}$ over 4.28–25.0 meV, a factor 5.8, gives $\sigma = 0.01440$ in all three cases, identical to five decimals, with κ identical and T_c scaling linearly. So the invariance holds numerically, not only in principle.

5. The learnable component

Section 4 measured what no model on $(\lambda, \omega_{\log}, \omega_2)$ can recover. This section measures what Allen–Dynes leaves on the table *within* those three inputs — error that a better function of the same arguments could absorb.

5.1 The raw discrepancy is not a scale-free question

Allen–Dynes error is measured here as $\varepsilon = \ln(T_c^{\text{ME}}/T_c^{\text{AD}})$. This has a structural defect which has to be removed before the error can be attributed to anything.

Both temperatures sit in the exponential denominator $\lambda - \mu^*(1 + 0.62\lambda)$. As λ falls toward $\mu^*/(1 - 0.62\mu^*)$ that denominator collapses and both T_c values fall, at different rates. So $|\varepsilon|$ *must* grow at low λ whatever the formula is doing — the growth is a property of the parameterisation, not of the approximation. Any analysis that reads a λ trend off ε directly is reading amplification.

The size of the effect is not marginal. Across λ tertiles, the median $|\varepsilon|$ falls by $5.23\times$ from the low to the high tertile — and the solver’s own sensitivity to μ^* , $|S_{\text{ME}}| = \partial \ln T_c / \partial \mu^*$, falls by essentially the same factor over the same range.

5.2 The sensitivity normalisation

We ask a scale-free question instead: *what shift in μ^* would account for this discrepancy?*

$$\Delta\mu_{\text{equiv}}^* = \frac{\varepsilon}{|S_{\text{ME}}|}$$

Two points that are easy to get wrong. The derivative must be the **solver’s**, not the closed form’s — $|S_{\text{AD}}|/|S_{\text{ME}}| \approx 1.9$ across this dataset, so using the wrong one leaves a large residual trend. And no additional solver runs are required: the dataset already contains T_c^{ME} at $\mu_{\text{ME}}^* = 0.1293$ and 0.1840 , so the finite difference is already present.

5.3 The λ trend is amplification; the shape trend is not

	raw $ \varepsilon $	normalised $ \Delta\mu_{\text{equiv}}^* $
$\rho(\cdot, \lambda \mid r)$	−0.639	− 0.039
$\rho(\cdot, r \mid \lambda)$	+0.400	+ 0.451
low/high λ tertile ratio	$5.23\times$	0.92 $\times$

Normalising removes the λ dependence of the error *magnitude*: the partial correlation collapses from −0.639 to −0.039, and the tertile ratio from $5.23\times$ to $0.92\times$. The apparent breakdown of Allen–Dynes at weak coupling is a property of the exponential denominator, not a failure of the fit.

Two qualifications, because the headline number is more estimator-dependent than it looks. First, −0.039 is the rank partial correlation — the standard formula on Spearman coefficients, equivalently residualising *ranks*. Residualising the raw values instead and then ranking gives +0.189, and the Pearson partial gives +0.248. The three are different statistics sharing a name, and only the first supports the word “completely”. Second, and independently of estimator choice, the *signed* residual retains λ structure that a magnitude correlation cannot see: a quadratic in $1/\lambda$ removes 17.9% of $\text{sd}(\Delta\mu_{\text{equiv}}^*)$ under ten-fold cross-validation (§5.6). The defensible claim is that normalisation removes the monotone λ dependence of $|\varepsilon|$, not that it leaves no λ dependence at all.

The shape dependence does the opposite. It survives normalisation and strengthens slightly:

$$\rho(r, |\Delta\mu_{\text{equiv}}^*|) = +0.450 \quad (p = 8 \times 10^{-27}), \quad \text{Pearson } +0.643 \quad (p = 7 \times 10^{-61})$$

on $n = 510$. Allen–Dynes handles the spectral shape parameter imperfectly, and the residual is predictable from r — which the formula already receives as an input, through f_2 . That makes this the *learnable* half by construction: a better function of the same three moments can absorb it.

The Pearson figure invites one misreading that we head off here: 0.643^2 is 41% of the variance **of the magnitude** $|\Delta\mu_{\text{equiv}}^*|$, which is not 41% of the error. Regressing the signed quantity on r removes 9.5% of its standard deviation. Both numbers are correct and they answer different questions; §5.6 reports the variance-scale version throughout, because that is the scale on which a headroom comparison would be made.

5.4 The reporting threshold moves the sign, not only the size

Every error statistic in this section is conditional on the cut that defines a superconductor. That cut moves the magnitude by $2.1\times$ and, more seriously, reverses the direction of the mean bias:

cut	n	median $ \varepsilon $	mean ε	excess kurtosis
all defined	583	0.0432	+0.0709	6.20
$T_c^{\text{ME}} > 0.05 \text{ K}$	520	0.0355	+0.0341	6.34
$T_c^{\text{ME}} > 0.5 \text{ K}$	369	0.0235	−0.0040	13.55
$T_c^{\text{ME}} > 1.0 \text{ K}$	303	0.0205	−0.0137	1.70

Allen–Dynes under-predicts on the full set and over-predicts above 1 K. **No directional statement about Allen–Dynes is meaningful without its cut**, and we state ours ($T_c^{\text{ME}} > 0.05$ K) with every figure.

The convention is a second, independent sign lever. Re-evaluating both T_c values at $\mu_{\text{AD}}^* = 0.13$ rather than 0.10, at **fixed** population ($T_c^{\text{ME}} > 1$ K, $n = 303$), moves the mean ε from -0.0137 to $+0.0374$ and the median $|\varepsilon|$ by $1.59\times$ (RMS $2.07\times$; on the $T_c^{\text{ME}} > 0.05$ K population the factors are $2.65\times$ and $3.26\times$). Threshold and convention therefore each reverse the sign of the mean bias on their own, and neither statement is conditional on the other.

The choice of functional matters comparably. On the identical $n = 520$ sample, the median $|\varepsilon|$ is 0.0355 and the RMS is 0.0993 — a factor 2.80 — because ε is heavy-tailed ($\kappa = 6.3$). Functional and threshold together span 0.0205 to 0.166, a factor of eight. This is why §4.7 requires the functional and the threshold to be declared before any ratio is formed: the numerator of such a ratio is a choice before it is a measurement, and §6.3 tabulates how much the choice is worth.

Within the second-moment family the choice is narrower but not innocent, and it interacts with the sign lever above. Root-mean-square and standard deviation are not the same second moment:

$$\text{RMS}^2 = \bar{\varepsilon}^2 + \text{sd}^2 \frac{n-1}{n},$$

an identity we verify to machine precision on every cut below. The RMS therefore carries the systematic bias $\bar{\varepsilon}$ and the standard deviation does not, so the two diverge exactly where Allen–Dynes is most biased:

cut	n	mean ε	sd	RMS	RMS/sd
all defined	583	+0.0709	0.1502	0.1660	1.105
$T_c^{\text{ME}} > 0.05$ K	520	+0.0341	0.0933	0.0993	1.064
$T_c^{\text{ME}} > 1.0$ K	303	-0.0137	0.0352	0.0377	1.072

The inflation is 6–11%. It is not a fixed overhead, because $\bar{\varepsilon}$ is itself convention-dependent: on the declared population held fixed across conventions ($n = 510$, all four T_c values positive and finite), the mean moves from $+0.0278$ at $\mu_{\text{AD}}^* = 0.10$ to $+0.1696$ at 0.13, a factor 6.1. The RMS inherits that dependence and the standard deviation does not: across the same two conventions the RMS grows $3.26\times$ while the standard deviation grows $2.72\times$ (on the $T_c^{\text{ME}} > 1$ K sample, $2.07\times$ against $1.95\times$, with the mean reversing sign). A second moment that moves with the convention by more than its own spread does is not the convention-free comparand it appears to be.

5.5 Two shape effects

Both halves of this paper are now about spectral shape: the learnable component is f_2 mishandling r , and the irreducible component is T_c variation among spectra sharing r . They are distinguished by construction rather than by inference, and they do not sum — §6.1 states both points, together with the conditional-expectation caveat that makes the second one necessary.

5.6 How much is learnable, measured as variance rather than correlation

Sections 5.3 and 5.4 report correlations and ratios of summary statistics. Neither answers the question a headroom comparison actually asks: *if a better closed form of the same three moments were fitted, how much of the Allen–Dynes error would remain?* That question has a direct answer — refit and measure the residual — and we give it here because §6.3’s first precondition turns on it.

The protocol is a recalibration of Allen–Dynes by ordinary least squares on $\varepsilon = \ln(T_c^{\text{ME}}/T_c^{\text{AD}})$ against functions of the moments the formula already receives, scored by ten-fold cross-validation (40 random fold assignments, same folds across designs) so that the reduction is out-of-sample. The population is the declared one, $T_c^{\text{ME}} > 0.05$ K, $n = 520$; $\text{sd}(\varepsilon) = 0.0933$.

recalibration class	CV sd	reduction
constant (intercept only)	0.0935	0.0%
linear in $1/\lambda$	0.0683	27.0%
quadratic in $1/\lambda$	0.0530	43.3%
quadratic in r	0.0917	1.9%
quadratic in $(1/\lambda, r)$	0.0521	44.2%
+ cross term r/λ	0.0514	45.0%

In-sample and cross-validated values agree to within 3% at every design, so the reduction is not overfitting. Adding $\ln \omega_{\log}$ to the largest design changes nothing (coefficient -0.0036 , bootstrap 95% CI $[-0.0124, +0.0046]$), which is the scale-invariance of §2.1 recovered as an empirical null on a quantity that was free to violate it.

Three readings follow.

The incumbent’s error and its class’s error are different quantities. Nearly half of $\text{sd}(\varepsilon)$ is removable by a six-parameter refit in variables Allen–Dynes already has. Any headroom statement must say which of the two it uses as its numerator; they differ by a factor 1.82, which is comparable to the functional lever ($2.80\times$) and the convention lever ($1.59\text{--}3.26\times$) of §5.4. This is a fourth choice on the same footing as §6.3’s three, and it moves the answer as much as they do.

Most of that reduction comes from the parameterisation rather than the shape. A quadratic in $1/\lambda$ alone reaches 43.3%; adding r takes it to 44.2%. On the raw scale r is worth about one percentage point of standard deviation. This is not in tension with §5.3 — it is §5.1’s amplification argument stated in variance terms rather than in tertile ratios, and it is why the normalisation of §5.2 exists. On the normalised scale the ordering reverses in relative terms: a quadratic in $1/\lambda$ removes 17.9% of $\text{sd}(\Delta\mu_{\text{equiv}}^*)$ and a quadratic in r removes 7.7%, with 22.8% jointly. Each is a mean over 200 random fold assignments and reproduces to 0.1 percentage point across three independent seeds.

So the shape effect is real but modest — eight percent of a standard deviation, which is a smaller quantity than $\rho = +0.45$ tends to suggest. What we claim for the r dependence is that it is present, robust, and of the same sign in both halves of the decomposition. We do not claim it dominates the Allen–Dynes error budget; the exponential denominator does that, and a two-parameter form in $1/\lambda$ alone removes 43% of the standard deviation.

0.0514 should not be read as the best achievable three-moment error. It is the residual of one declared class, and a richer class would go lower. The sequence of such residuals is what §4’s floor bounds from below, and the floor is the only member of that sequence which does not depend on the class.

6. Discussion

6.1 What the two halves say together

The learnable component is Allen–Dynes mishandling the shape parameter r that it already receives: after the parameterisation artifact is removed, the residual correlates with r at $\rho = +0.45$ over 510 materials and shows no residual monotone λ dependence in magnitude (-0.039). Its size should be stated alongside its significance: on the normalised scale a quadratic in r removes 7.7% of the standard deviation, against 17.9% for a quadratic in $1/\lambda$ (§5.6). The r effect is robust and it is modest; the claim of this paper is the first, not the second. The irreducible component is the spread of T_c among spectra sharing all three moments, and it too rises with r — $\rho = +0.92$ across nine cells, robust to leave-one-out, to equalising sample size, and to fixing donor composition.

Both halves are therefore about spectral shape, which invites the objection that they are the same effect counted twice. They are not, and the distinction is by construction rather than inference: the learnable part is variation *predictable from r* , an input the formula already has; the irreducible part is variation *at identical*

r and identical λ , $\omega_{\log}$ — information destroyed at the moment of compression, which no function of those three numbers can recover. They are measured on disjoint axes — one across materials at differing moments, one across spectra at fixed moments. That both scale with the same parameter is the physical content of the result.

The two components are not, however, two halves of a total. Their orthogonality requires a common conditional expectation, and §4.3’s is taken over a constructed ensemble roughly threefold narrower than the physical one (§4.8). The decomposition $\varepsilon = [T_c^{\text{AD}}(m) - \mathbb{E}(T_c^{\text{ME}} | m)] + [T_c^{\text{ME}} - \mathbb{E}(T_c^{\text{ME}} | m)]$ is orthogonal by definition, but only if both terms use the *same* conditional expectation, and these do not. We therefore report the irreducible term as the bracket $0.0085 \lesssim \sigma \lesssim 0.029$ rather than as a complement to the learnable term, and note that both of §4’s correlations reproduce on near-moment-matched *real* materials — a factor 1.64 between $r < 1.2$ and $r \geq 1.3$ — so neither is an artifact of the construction.

In practical terms $\omega_2/\omega_{\log}$ is where the three-moment representation is weakest in both senses at once — it is the argument the fit handles least well, and the argument along which the compression discards the most. Anyone looking for a fourth descriptor would presumably want a shape statistic orthogonal to r rather than a refinement of λ , though we have not tested that.

6.2 The limitation we regard as central

Convexity of the moment-matched set is well posed; a measure on that set is not. Every number in §4 is conditional on the choice of ensemble, and we have shown that the choice moves the answer materially: the cross-cell dynamic range is $23.1\times$ under each target’s own reachable donor set and $14.1\times$ under a fixed common set, a gap of roughly 40%.

The constraint is **directional rather than general**, and saying so precisely matters. Reweighting every target’s set to the *pool* density is feasible at the two lowest- r cells — no pool mass uncovered, effective sample sizes 746 and 666 — and fails at the other seven, because at high r the low- r donors are absent rather than merely rare and no importance weight can manufacture a spectrum the pool does not contain. Reweighting instead to the *narrowest* cell’s donor- r density is feasible at all nine, with effective sample sizes 207–266. That is a better control than the common-set restriction reported in §4.6: it exceeds the restriction’s $n = 193$ at every cell, keeps each cell’s full natural set with weights rather than discarding donors, and matches the whole density rather than only the support.

What no reweighting escapes is the choice itself. Escaping it requires a generative model of $\alpha^2 F$ that can *create* support rather than reweight it — the natural next construction, whose own measure would then have to be declared in turn.

There is, then, no measure-free irreducible floor. We take that as a reason to quote the floor with its measure stated and its sensitivity tabulated, rather than as a reason not to quote one; §6.3 does the first.

6.3 Headroom, and what it is conditional on

Dividing the Allen–Dynes error by the floor requires three things to be fixed first: the recalibration class (how rich a function of the same three moments the numerator is allowed to be — the trivial member of that family is “report the standard deviation and the bias separately”, §4.7), the population, and the μ^* convention.

The declared value

Under $T_c^{\text{ME}} > 0.05$ K, $\mu_{\text{AD}}^* = 0.10$, and the trivial recalibration class, against the nine-cell median floor $\hat{\sigma} = 0.00847$:

$$\text{headroom} = \frac{0.0933}{0.00847} = 11.0$$

Allen–Dynes leaves roughly an order of magnitude between where it is and where the three-moment representation stops being able to help. How much of that order of magnitude is real depends on the three choices, as

follows.

variant	numerator	denominator	ratio	vs declared
declared	0.0933	0.00847	11.0	1.00×
class: RMS (bias folded in)	0.0993	0.00847	11.7	1.06×
class: six-parameter refit (§5.6)	0.0514	0.00847	6.1	0.55×
population: $T_c^{\text{AD}} > 1$ K	0.0352	0.00847	4.2	0.38×
convention: $\mu_{\text{AD}}^* = 0.13$	0.1747	0.00847	20.6	1.87×
denominator population coverage	0.0933	—	5.4–10.0	0.42–0.99×, signed
denominator: §4.8 upper bracket	0.0933	0.0280	3.3	0.30×

Across these alternatives the ratio runs from 3.3 to 20.6, a span of $6.2\times$ about the declared 11.0. So the order of magnitude survives the choices; the leading digit does not, and we would not defend it. The coverage row is a different kind of entry from the others — it is a correction rather than an alternative convention, and it moves the ratio down. The subsection below works it out and folds it in, giving 5–11. Readers who prefer other declarations can read off their own value.

Headroom as a function of shape

The floor spans $23\times$ across the nine cells (§4.6) and the Allen–Dynes error does not span anything like that, so a single ratio conceals most of the structure. Restricting the material population to a $\pm 6\%$ window in r about each cell — so that numerator and denominator describe the same region of shape space — gives:

cell	r	floor	n_{mat}	AD sd	headroom	recalibrated
CrRh ₃	1.057	0.00202	81	0.0539	26.7	14.2
HfRu ₂ Ta	1.070	0.00105	104	0.0660	63.0	30.6
CoTi	1.161	0.00315	188	0.0937	29.7	15.7
AsZr	1.213	0.00320	190	0.1005	31.4	16.7
AlHfNi ₂	1.303	0.00955	150	0.0907	9.5	5.3
BeNi ₂ Si	1.333	0.01844	137	0.1036	5.6	3.0
Ga ₂ LiRu	1.410	0.00847	99	0.1016	12.0	6.3
Be ₂ PdZn	1.483	0.02421	73	0.1054	4.4	2.2
Se ₂ V	1.561	0.01987	55	0.0583	2.9	2.1

$$\rho(r, \text{headroom}) = -0.833, \quad p = 0.005, \quad n = 9.$$

The low- r end of this range can be treated exactly. At $r = 1$ the moment-matched set collapses to the Einstein spectrum (§4.1), so $\sigma = 0$ by Jensen and the power-mean inequality. The numerator does not vanish along with it, because materials near $r = 1$ still differ in λ and the Allen–Dynes error retains a spread on that account. Headroom therefore diverges as $r \rightarrow 1$. The accompanying statement is not something we measured:

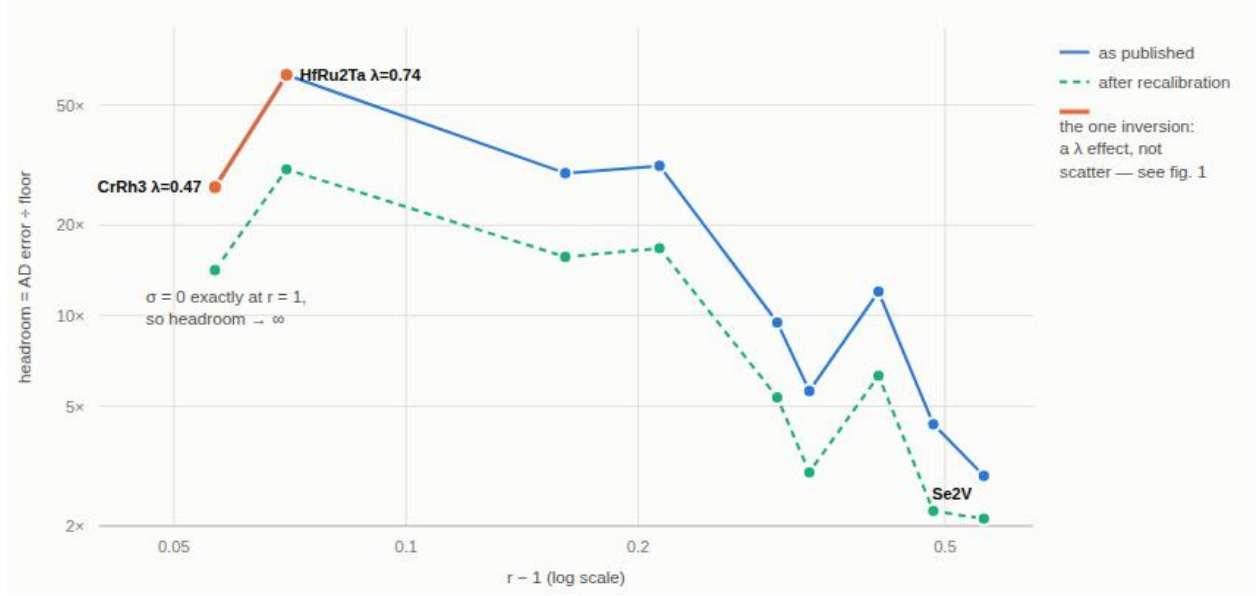


Figure 2: Headroom against $r - 1$ on log-log, numerator and denominator matched in shape space.

At $r = 1$ the entire Allen–Dynes error is in principle learnable, because the three moments determine the spectrum uniquely.

This strengthens the case against the scalar past where §6.3 had taken it. The ratio 11.0 does not merely average over a $23\times$ spread; it averages over a quantity that is unbounded at one end of its range.

The table contains one inversion. CrRh₃ and HfRu₂Ta sit at almost the same shape ($r = 1.057$ and 1.070) yet their headroom differs by $2.4\times$, which at first looks like noise in a $\rho = -0.833$ relation. It is a coupling effect: both cells lie below §4.9’s minimum, on the branch where the floor rises as λ falls, and they differ in λ by 0.27 . Reconstructing the nine cells’ floor *ratios* from §4.9’s ladder slope together with §4.8’s shape exponent — a slope measured on synthetic targets at $r = 1.19$, applied to real-material cells at $r \approx 1.06$ — predicts all 21 pairs below the minimum with a median error of 13% in the ratio, against a factor 2.7 for a model using the coupling alone. §4.9 and the headroom table were built independently of each other, so the agreement is a cross-check rather than a restatement.

It also bears on the row a referee is most likely to attack. HfRu₂Ta’s 63.0 rests on the smallest floor in the grid, with $\kappa = 26.5$ and the worst precision of the nine at 9.7%. The λ curve predicts it anyway, which is some protection against its being an artifact of that single cell.

Headroom falls by roughly an order of magnitude across the observed range of r . The two columns that build it show why: the floor rises steeply with r ($\rho = +0.917$) while the Allen–Dynes error rises only weakly ($\rho = +0.467$), so the denominator gains on the numerator as r grows.

The trend is not an artifact of the matching window. Widening it from $\pm 4\%$ to $\pm 15\%$ moves the median ratio only between 11.3 and 13.1 and strengthens the correlation monotonically ($\rho = -0.817$ to -0.917); leave-one-out over the nine cells leaves ρ between -0.762 and -0.905 , with $p \leq 0.028$ at every deletion.

We would rather the result were read in this form than as the scalar. For a near-Einstein spectrum the three-moment representation is nowhere near its limit, and a better closed form has a great deal to give. For a broad, structured spectrum it is already close to exhausted, and most of what remains is information the compression destroyed. Averaging the two regimes together gives a number that describes neither of them.

The population of the targets is a coverage question, not a measurement one

The nine floor cells were selected from materials with $T_c^{\text{AD}} > 1 \text{ K}$ — a cost bound — while the numerator is quoted on $T_c^{\text{ME}} > 0.05 \text{ K}$. This looked like an outstanding precondition. It is a smaller thing than that, for a reason worth stating explicitly: σ is a function of a moment triple, not of a population. It is measured by tilting all 806 donors onto the target moments, so the cut from which a target was drawn does not enter its value. The nine measured floors are correct at their nine triples. What the cut affects is whether those triples are representative of the declared population — coverage, not correctness.

Re-selecting on the declared population changes all nine targets (the tertile boundaries move, so there is zero overlap), which makes a cell-by-cell comparison impossible; but it lets the question be asked the right way, by comparing the two populations' coverage of the fitted floor surface:

	n	median λ	r at 10 / 50 / 90 %	inside cells' λ	inside cells' r
$T_c^{\text{AD}} > 1 \text{ K}$	304	0.612	1.081 / 1.253 / 1.628	86.2%	84.9%
$T_c^{\text{ME}} > 0.05 \text{ K}$	520	0.447	1.086 / 1.244 / 1.579	54.2%	86.7%

The r distributions are nearly identical; the λ distributions are not. That asymmetry decides which of the results is exposed to the selection cut:

- The **per-cell result** — the r dependence of headroom, and the r trend of §4.4 beneath it — is not affected. Both populations place the same fraction of their mass inside the cells' r range, to within two points, and their r quantiles agree to better than 3%.
- The **scalar** is affected, through λ only. Nearly half the declared population lies below the cells' lowest λ .

The correction can be signed, and it runs against the headline number. Note first that it cannot be obtained from the cross-cell fit $\ln \sigma = a + p \ln(r - 1) + b\lambda$: that fit's λ coefficient is confounded with r (§4.9), so it does not estimate the derivative this argument needs.

The sign follows from §4.9's curve and the cells' placement on it, and it does not require knowing which side of the coverage gap dominates. The nine cells span $\lambda = 0.411$ to 1.250 and therefore **straddle the minimum at $\lambda \approx 0.90$** . The 238 materials outside that range divide 216 below and 22 above; those below sit on the descending limb, those above on the ascending one, and **both limbs are higher than the middle**. Evaluating §4.9's curve at each material's own coupling, 100% of the materials in *each* tail have a larger floor than the median cell. The cells therefore sit where the floor is smaller than a representative sample would give, the denominator of the scalar is too small, and **11.0 overestimates headroom**.

With the ladder now extending to $\lambda = 0.40$, the declared population's median coupling lies **inside** the measured range and the magnitude no longer rests on an extrapolated step:

$$\sigma(0.742) = 0.0038 \longrightarrow \sigma(0.447) = 0.0091,$$

a factor 2.4 higher at the declared population's coupling.

Converting that into a correction on the scalar requires choosing how each population is summarised, and the choice is not settled by the measurement. Two things pull against each other. The cells sit at *higher* coupling than the declared population (median λ 0.742 against 0.447), which on the descending limb makes their floors **smaller**; but they also sit at higher shape (median r 1.303 against 1.244) by tertile construction, which makes their floors **larger**. Whether the second is treated as part of the coverage gap decides most of the answer:

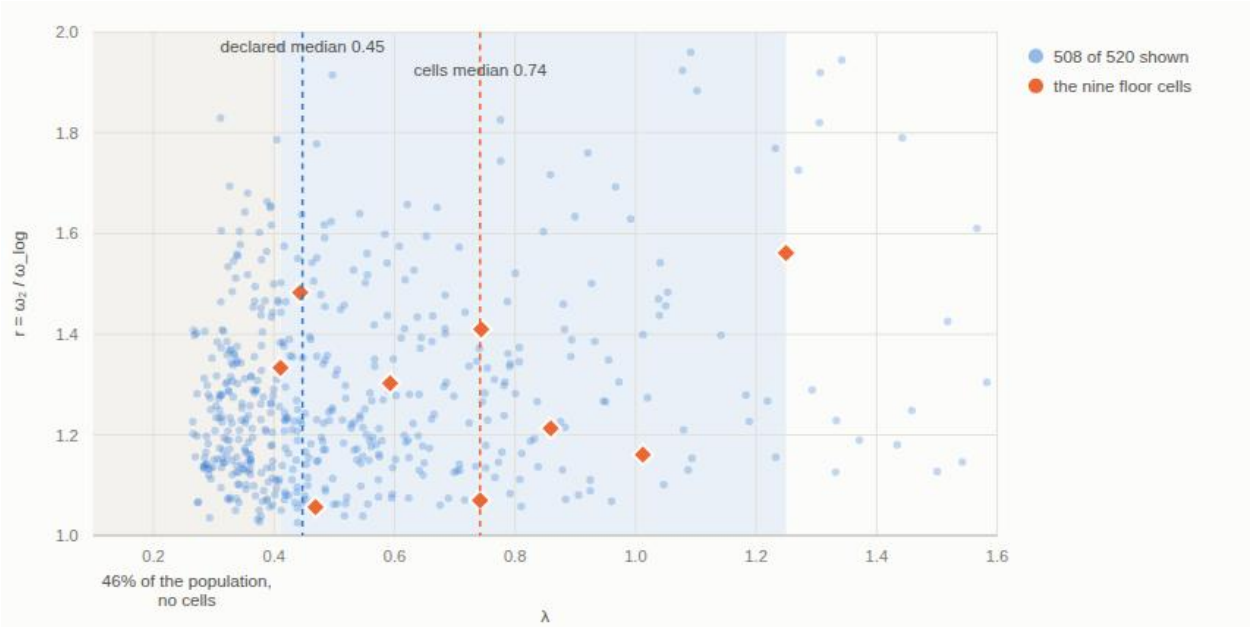


Figure 3: The (λ, r) plane: 508 of the 520 declared-population materials with the nine floor cells overlaid.

estimand	correction	corrected headroom
σ at each population's median λ	0.42	4.6
median per-material prediction, λ only	0.61	6.8
median per-material prediction, joint (λ, r)	0.68 – 0.99	7.5 – 10.9
cells' <i>measured</i> median floor over declared prediction	0.82	9.0

We therefore quote

$$\text{coverage-corrected headroom} \approx 5\text{--}11,$$

against the uncorrected 11.0 — a correction whose *direction* is established (§4.9 and the tail argument above: 100% of the uncovered materials on both sides have larger floors than the median cell) but whose *magnitude* the estimand choice moves by a factor of two. The upper end is additionally a bound rather than an estimate: 38.8% of the declared population still sits below the lowest rung and is clipped to it, which understates those materials' floors and so overstates the correction. Closing that would need a rung near $\lambda = 0.30$, whose cost we measured at two orders of magnitude above the reference rung (§4.9) — not worth spending to tighten a bound that already errs in the safe direction.

We have folded this into the headline figure rather than printing it alongside one. Leaving 11.0 in place with the correction quoted separately would leave the reader to do a multiplication we have already done.

The residual caveat is the one the table makes visible: the nine cells jointly cover 47.3% of the declared population in (λ, r) , against 75.0% of the population they were drawn from, and the shortfall is entirely at low λ . The scalar headroom of 11.0 is therefore a statement about the covered region. The r -resolved table is not subject to this, which is one more reason to prefer it to the scalar.

6.4 Limitations

Reference standard. The comparison is against DFT, not experiment; §1 states this and §3.5 quantifies it. The two DFT pipelines we compared disagree by 31% in λ for the same material while agreeing to 2% in r — a controlled smearing sweep on a single pipeline gives the same asymmetry, 44% against 0.59%, because λ is linear in the amplitude of $\alpha^2 F$ and r is invariant to it. Conclusions expressed in r are therefore substantially more robust to DFT convergence error than conclusions expressed in λ , which is fortunate given that the surviving result is an r result.

Nine cells. All cross-cell statistics rest on nine points. The r trend survives every control we applied. The coupling dependence does not resolve *on this grid* ($\rho(\sigma, \lambda \mid r) = -0.652$, $p = 0.080$, and a single deletion takes it to $p = 0.377$), and no quantity of additional donors would change that, since it is a limitation of the grid’s design rather than its sampling. §4.9 resolves it by a different route — synthetic targets at fixed r — and finds a coefficient of the opposite sign to the cross-cell fit, which is why we report no value for the latter.

The tail. We have no explanation for it. $\ln T_c$ over moment-matched spectra is leptokurtic at every cell, with κ from 2.7 to 26.5, and κ falls as r rises. We report this as a measured property. A composition-based explanation is ruled out — two cells sharing 98% of their donors differ by $2.5\times$ in κ — and we do not currently have a mechanism. It matters beyond bookkeeping: σ is a second moment on a heavy tail, so any generative model intended to replace the empirical ensemble must be validated against the induced $\ln T_c$ tail and not merely against bulk shape statistics, or it will reproduce the mean and miss the floor.

Isotropic, constant-DOS. The solver treats neither anisotropy, a variable density of states, vertex corrections, nor the frequency dependence of the screened Coulomb interaction (§3.7).

6.5 Outlook

Three things would materially advance this work, in order of what they settle per unit of effort.

First, **a mechanism for the low- λ rise.** §4.9 now maps $\sigma(\lambda)$ at fixed r over $0.40 \leq \lambda \leq 20$ with two turning points. The strong-coupling fall is explained — the asymptote depends only on matched moments. The weak-coupling rise is measured but not derived; the exponential-denominator argument of §5.1 predicts its sign and not its slope, and a quantitative account would say whether σ diverges at the Migdal–Eliashberg threshold or turns over again.

Second, **a generative model of $\alpha^2 F$** fitted to the observed spectra, which is the only construction that escapes the support constraint of §6.2 and the only route to sample sizes large enough for the heavy tails of §4.3. Its validation criterion should be the induced T_c tail.

Third, calibrated predictive uncertainty — a quantile-based predictor that reports intervals, tested for whether its residual uncertainty converges to the floor measured here. One caution: a quantile-scored model needs a quantile-valued floor, so the estimand has to be fixed before either is computed. Reporting σ and comparing it against interval coverage would reproduce the functional mismatch of §6.3 in a new place.

Data and code availability

All data and code required to reproduce every number, table and figure in this paper are public at github.com/abhinavsharma-ui/superconductivity-uncertainty.

The repository contains the Migdal–Eliashberg solver (`src/eliashberg.py`) and its ten internal checks (`src/verify_eliashberg.py`), the moment-matched ensemble generator (`src/spectral_generator.py`), the scripts that produce the nine-cell grid and the $\sigma(\lambda)$ ladder, and the verification scripts that regenerate the quantitative claims of §4–§6 directly from the stored outputs. Input $\alpha^2 F(\omega)$ are the released BETE-NET database [Gibson *et al.* 2025], redistributed unmodified.

Where a number in this paper is reported to a stated number of digits, the corresponding script asserts that value rather than printing it, so that a change in the underlying data fails loudly rather than silently.

Competing interests

The author declares no competing interests.

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Verification status. All references above except 1–4 were checked against the publisher record during preparation; refs. 1–4 are standard and were not re-checked. Two entries were corrected as a result of that check, and one further reference — a least-upper-bound result attributed to Leavens (1975) — was removed because we could not retrieve the primary source. The extremal statement it was cited for is given by ref. 8 (Gao *et al.*) in retrievable form.

12. Recent work bounding T_c under various constraints, not used quantitatively here but relevant to §4.1’s extremal boundary: *A bound on the superconducting transition temperature*, npj Quantum Mater. **3** (2018); *Fundamental limits on the electron-phonon coupling and superconducting T_c* , arXiv:2407.12922; *Upper bound on T_c in a strongly coupled electron-boson superconductor*, Phys. Rev. B (2025), arXiv:2505.02894.